Text, OCR, or Pixels?

A Retrieval Decision Matrix for Quant-Finance Documents, Built on Eighteen Years of the osQF Archive

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AssetFlow.ai https://github.com/travisjakel/osqf-archive

July 2026

Abstract

Much of the written record in quantitative finance is visually dense: broker research, filings, and conference slide decks are full of tables, charts, and equations that plain text extraction mangles or drops. We wanted to know, with numbers, which retrieval representation works for which kind of content. As a testbed we used the complete osQF/R-Finance archive: 626 talks from 2009 to 2025, fully public, and messy in the same ways practitioner document stores are. We built a local pipeline (resumable R stages, a 14B open-weight LLM doing forced tool-call extraction with verbatim evidence-quote validation, a DuckDB semantic index) and compared three retrieval layers under a vision-judged evaluation: native text extraction, VLM-OCR re-extraction behind an acceptance gate, and pixel-level page-image retrieval. Native text wins on born-digital documents. Gated OCR recovers image-rendered decks without regressing anything. Pixel retrieval answers chart questions on which both OCR models we tested score 0.00. We condense the results into a decision matrix by document class, and the whole system runs on one consumer GPU with no cloud calls in the data path. Code, index, a hosted MCP endpoint, and a browser demo are all public, so attendees can query eighteen years of this conference from their own tooling during the talk.

Keywords: retrieval-augmented generation, document AI, OCR, local LLMs, DuckDB, open source, quantitative finance

1 Introduction

The archived editions of this conference (R/Finance 2009–2019, osQF 2022–2025) have produced hundreds of slide decks and papers, publicly hosted at https://www.osqf.org/archive/, and effectively unsearchable. Finding “that talk where someone showed GARCH estimates change with the optimizer and starting values” means paging through year indexes and opening PDFs one at a time. The archive is a knowledge base without a query interface.

The same problem sits on every quantitative desk. Broker research, vendor decks, prospectuses, and filings fail in the same ways the osQF archive does: born-digital text next to image-rendered text, dense tables, charts whose content exists only in pixels, formulas, code snippets, and two decades of drifting file formats (including legacy binary `.ppt` that modern parsers refuse). A retrieval system that assumes clean text quietly drops the content practitioners care about most: the numbers in the exhibit.

We therefore treat the conference's own archive as a domain-matched, legally clean, fully public testbed, and ask a practical question: *for each kind of content, should we retrieve from extracted text, from VLM-OCR output, or from the raw page image?* Our contributions:

1. An open, reproducible pipeline that turns the complete osQF archive (626 talks, 2009–2025) into a structured, semantically searchable knowledge base with *verbatim evidence-quote validation*, running entirely on one consumer GPU (§2).
2. A quantified, vision-judged comparison of three retrieval layers: text, VLM-OCR (including a bake-off of two 2026 open-weight OCR models), and pixel-level page-image retrieval (§3).
3. A decision matrix mapping document classes to retrieval layers, with the evidence behind each cell (§4).
4. Associative (multi-hop) retrieval over the archive's talk–topic–speaker graph, implemented natively in R (§5).
5. Open artifacts: MIT-licensed code, the downloadable index, a hosted read-only MCP endpoint for any attendee's AI tooling, and a zero-install browser chat over the same tools (§7).

2 Corpus and Pipeline

The osQF archive pages for 2009–2025 (no conference was held 2020–2021) yield 835 agenda rows, which reduce to **626 real talks** after filtering event headers (breaks, lunches, panels). 469 talks have a retrievable file: 388 PDF, 63 PPTX, 8 legacy binary PPT, 1 PPTM, and 9 hosted HTML slide pages; the remaining file links are dead (pre-2020 hosting). After OCR repair (§3.2), 462 of the 469 are extracted and indexed; every count in this paragraph can be recomputed from the released manifest. The pipeline (Figure 1) is five resumable R stages; every stage is idempotent and cache-backed, so partial failures resume rather than restart.

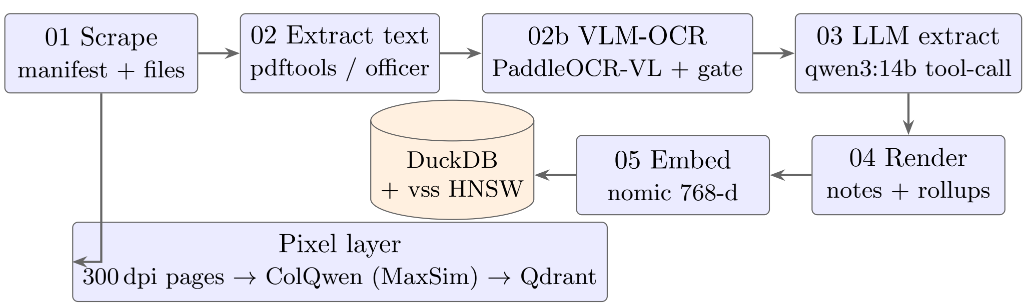

Figure 1: The five-stage text pipeline plus the parallel pixel layer. A single 300-dpi render of all 462 extractable decks (12,276 page images) feeds both the OCR repair stage and pixel retrieval.

Structured extraction with evidence quotes. Stage 03 sends each talk's text to a locally served 14B open-weight model (qwen3:14b via Ollama) with a *forced tool-call schema*: headline, topics, methods, datasets, code/tools referenced, key findings, applications, references. The critical design rule is **do not infer**: every key finding must carry an `evidence_quote` that is a verbatim substring of the source text (≤200 chars), validated post-hoc by whitespace-normalized substring matching. The corpus-wide evidence pass rate is **69.2%**, and because it is computed mechanically it doubles as a per-talk trust label on every rendered note. Failed validations stay visible in the rendered notes instead of being dropped. What are the failing ∼31%? A manual classification of a random sample of 40 failing quotes shows the dominant failure mode is *citation format, not fabrication*: 33 of 40 quote content that is present verbatim or near-verbatim in the source but stitched across fragments with ellipses or reformatted from tables (which the strict single-substring test correctly rejects), 2 are empty or sub-threshold quotes, and 5 cite table or figure values that never reached the extracted text — the one class that cannot be verified against the text layer at all. The classified sample ships with the code.

Semantic index. Rendered notes are chunked by section and embedded with nomic-embed-text (768-d) into DuckDB with the `vss` extension (HNSW, cosine): **4,111 chunks** across 462 talks, queryable with metadata filters (year, speaker, topic, method) in a single SQL expression. DuckDB is a deliberate choice over a vector-database service: the entire index, notes included, is a single ∼50 MB file that ships as a release asset.

Known-item retrieval evaluation. The index's core promise is "find that talk," so we measure it directly: 25 author-written known-item queries, each targeting one talk, paraphrased from the *source decks* rather than the indexed notes (limiting circularity) and stratified across years, file formats, and OCR-recovered decks (15/25). Semantic retrieval places the right talk at rank 1 for **68%** of queries (hit@3 0.76, hit@5 0.80, MRR 0.74). A BM25 full-text arm over the same index answers the same queries at hit@1 0.64 but hit@3 **0.84**: on known-item queries with distinctive vocabulary the two arms are complementary rather than ordered, which is why the release ships both (semantic default, BM25 as the zero-model fallback). The query set, gold labels, and harness are released with the code.

3 Three Retrieval Layers

3.1 Layer 1: Native text

`pdftools` and `officer` handle born-digital documents well and are the cheapest layer by orders of magnitude. They fail in three specific ways: image-rendered text and SmartArt in slide decks (thin extractions), garbled text layers in old PDFs, and legacy `.ppt` (unreadable). Before repair, 26 of the 469 retrieved files failed extraction outright and dozens more were thin or garbled.

3.2 Layer 2: VLM-OCR, with a judged bake-off

We converted all 74 PowerPoint decks to PDF (LibreOffice headless, closing the legacy-`.ppt` gap), rendered every extractable deck to 300-dpi page images, and re-extracted the corpus with an open-weight vision-language OCR model served by vLLM on the same GPU. Two 2026 open-weight candidates were bake-tested on 15 decks stratified across the known failure classes (garbled PDFs, legacy PPT, thin PPTX, zero-evidence extractions), with a frontier vision model judging 37 dense pages against the rendered page image (Table 1).

	Unlimited-OCR (3B MoE)	PaddleOCR-VL (0.9B)
Fidelity, all judged pages	0.48	**0.76**
Fidelity, chart pages	0.54	0.57
Pages with hallucinated content	13/37	**4/37**
Downstream evidence-quote pass	39.7%	**70.1%**
Model size (BF16)	6.4 GB	**2.0 GB**

Table 1: Vision-judged OCR bake-off on 15 stratified osQF decks. The larger, newer model loses on overall fidelity, hallucination rate, and downstream evidence pass; both models are near-useless on charts (§3.3). A practical finding: both models respond only to their trained task prompts ("`OCR:`", "`Free OCR.`"); free-form instructions return empty output.

After the initial bake-off we added the current OmniDocBench leader (MinerU2.5, layout-routed hybrid parsing [6]) as a third arm on the identical pages: fidelity tied the champion (0.74 vs. 0.76), hallucinations tied (3–3), and downstream evidence-quote pass trailed by 4.6pp, so the champion held. Leaderboard rank did not predict rank on our corpus, which is the argument for judging candidates on your own documents.

Two engineering guards matter more than the model choice. First, an *acceptance gate*: OCR output replaces the native text layer only if it is ≥300 characters *and* ≥0.7× the native extraction's length. 349 of 462 decks were accepted and 113 rejected (mostly 2009–2011 LaTeX beamer decks whose native text layer is better than any OCR). Second, a *best-of-both revert*: if downstream structured extraction fails on the OCR text where it previously succeeded, the deck reverts to its pre-OCR state. The re-extraction is therefore monotone: it recovered 19 previously dead talks (including seven of the eight legacy `.ppt` decks) and broke none of the existing ones. A caveat this design makes explicit: for OCR-sourced notes, evidence quotes validate against OCR output rather than source truth, so hallucinated OCR could silently "validate", which is why the bake-off weighted hallucination above raw fidelity.

3.3 Layer 3: Pixel retrieval

Charts defeat both OCR models (fidelity ≈0.55, and frequently 0.00 on pure-plot pages). The pixel layer skips transcription entirely: all **12,276 page images** are embedded directly into a Qdrant collection; text queries embed into the same space, and a small local VLM (Qwen2.5-VL-3B) reads the retrieved page image to produce an answer. Retrieval began as single-vector page embeddings (Qwen3-VL-Embedding-2B, 2048-d) and was subsequently migrated to *late-interaction* multi-vector retrieval (ColQwen2.5: MaxSim over sequence-pooled patch embeddings, 128-d per vector), following the ColPali design [2]. In a judged-relevance evaluation (12 queries; a frontier vision model grading each retrieved page), late interaction lifts precision@1 from 0.75 to **0.83** and precision@3 from 0.67 to **0.75**. The probe set is small—the P@1 gain is one additional query resolved at rank 1—so we read these as directional; the clearest win is qualitative: "model-selection tables with AIC/BIC columns" goes from 0/3 relevant pages retrieved to 3/3, the fine-grained-detail regime where the ColPali line of work predicts multi-vector scoring should help. In an earlier controlled A/B on a separate corpus of research-note PDFs (not part of this release), pixel retrieval beat text retrieval by **+0.50** accuracy on table questions while tying on prose; we cannot redistribute that corpus, so treat the number as corroboration. On this corpus the effect is starker. For a chart question about a 2011 talk's oil-price/housing scatter plot (a page where both OCR models scored 0.00 fidelity), pixel retrieval returns the correct pages at ranks 1–3, and the local reader describes the axes, ranges, and the quadratic fit correctly. That content never existed in any text layer.

4 The Decision Matrix

Content / document class	**Layer**	**Evidence**
Born-digital text (filings, papers, beamer decks)	Native text	113/462 decks: native beats OCR; zero cost
Image-rendered text, SmartArt, scans, legacy formats	VLM-OCR + acceptance gate	349/462 accepted; 19 dead talks recovered; gate blocks regressions
Tables (broker grids, exhibits)	OCR if gated; pixels when values must be exact	OCR-VL 0.76 fidelity; pixel +0.50 vs. text on table Q&A in an earlier A/B (§3.3)
Charts and plots	Pixels only	both OCR models ≈0.55 on chart pages, 0.00 on pure plots; pixel retrieves rank-1
Formulas	VLM-OCR (LaTeX output)	supported by both models; validate downstream
Cross-document "who/what relates to X"	Graph + PPR (§5)	worked query in Table 3; 322 of 462 talks have ranked neighbors

Table 2: Which representation to retrieve from, by content class. The mapping maps naturally onto practitioner stores: broker research is the PPTX column, filings are the born-digital column, vendor chart books are the pixel column.

The pattern across every row: *extraction gets lossier as pages get more visual, and the loss is silent*. Rather than hunt for one perfect extractor, we serve each document from the cheapest layer that preserves its content, and let mechanical gates (length ratios, evidence-quote validation, judged fidelity) decide per document when to escalate.

5 Associative Retrieval: the Archive as a Graph

Semantic search answers "find talks about X." It cannot answer "who, across eighteen years, is the person to ask about X?", because that association routes through entities rather than text similarity. We build a talk–entity graph (462 talks; 985 nodes; 890 edges over shared topics, methods, and speakers) and run Personalized PageRank from cosine-seeded talks, natively in R via `igraph`. Three design choices, found the hard way, that a sparse curated graph needs: entity-only edges (dense kNN edges collapse PPR into degree centrality), IDF-weighted edges ($w = \log N/\text{deg}$, so one shared rare topic outweighs a hub speaker), and ranking by PPR *lift* over the unpersonalized baseline (de-emphasizing celebrities). Queries like "robust portfolio optimization" then surface the archive's actual specialists, the researchers who return to the topic across editions, rather than simply the most prolific speakers. Table 3 shows that query's output verbatim. All eight results sit *outside* the plain cosine top-10 — the associative layer surfaced them through the entity graph — one speaker returns to the topic across four editions, and the archive's most prolific speaker (12 talks) is correctly absent.

Rank	Talk	Year	PPR lift
1	Risk Parity Portfolios with `riskParityPortfolio` (Palomar)	2019	9.3
2	Portfolio Selection with Multiple Criteria Objectives (Pfaff)	2016	8.8
3	Scenario Analysis of Risk Parity using RcppParallel (Foster)	2017	7.8
4–7	Portfolio Allocation with Cluster Risk Parity + three later editions (Kapler)	2013–19	4.7
8	Portfolio Optimization: Utility, Computation, Equities (Burkett)	2014	4.6

Table 3: Associative search output for "robust portfolio optimization" (top 8 by PPR lift over the unpersonalized baseline; cosine seeds excluded). None of these appear in the bare cosine top-10.

6 Hardware and Cost

Every stage—scraping, extraction, the OCR bake-off and full re-extraction, embedding, judging setup, and serving—ran on a single RTX 3090 Ti (24 GB) under WSL2, with open-weight models and zero cloud calls in the data path.[1] Approximate wall-clock: full-corpus render 35 minutes (CPU); full-corpus OCR ∼4 hours; structured re-extraction ∼2.5 hours; pixel embedding ∼3 hours; the semantic index rebuilds incrementally in minutes. The GPU is the only scarce resource, and stages are sequenced so no two model servers contend for it. A desk could run the identical stack against its own document store without any data leaving the building. For broker research and internal documents, that is a requirement, not an optimization.

7 Reproducibility and Access

All artifacts accompany this paper:

- **Code** (MIT): the full pipeline, evaluation scripts, and MCP server at `https://github.com/travisjakel/osqf-archive`.
- **Index**: a single self-contained DuckDB release asset (∼50 MB) holding the semantic index, talk manifest, derived notes, tag tables, precomputed expert/related-talk graphs, and a persisted full-text index. It works offline with zero model downloads via the built-in BM25 fallback, or with full semantic search after one `ollama pull`.
- **Hosted MCP endpoint, live now**: a read-only Model Context Protocol server over the complete archive (semantic search, talk notes, expert-finding, guarded SQL) at `https://osqf.assetflow.ai`, so attendees can point Claude, or any MCP client, at eighteen years of this conference during the talk. The shared attendee token appears on the talk's closing slide; a reviewer token accompanies this submission. Client setup: `osqf_mcp/CLIENT_SETUP.md` in the repository.
- **Browser demo**: for anyone without an MCP client, `https://osqf.assetflow.ai/chat` wraps the same seven tools in a minimal agent loop behind a one-page chat UI (a small cloud chat model, budget-capped and token-gated; the archive itself stays local). Answers cite talks with links back to the osqf.org sources, and each reply displays the tool calls it made, so what the room sees is the same tool sequence an MCP client would run.

[1] Cloud use is confined to two optional edges outside the data path: the bake-off *judge* (a frontier vision model grading OCR transcripts against page images—acceptable because the corpus is public; the equivalent local judge, Qwen2.5-VL, is included in the release for private corpora) and the browser demo's chat model (§7). Extraction, embedding, and indexing never leave the machine.

- **Provenance**: original speaker PDFs are *not* redistributed; notes carry short validated quotes and link to `osqf.org` sources. Model weights are pulled from their upstream repositories; licenses are catalogued in `NOTICE.md`.

8 Related Work and Conclusion

Pixel-level document retrieval follows the direction of ColPali [2]; our associative layer is a native-R adaptation of HippoRAG's PPR-over-entity-graph design [1]; the OCR candidates are PaddleOCR-VL [3] and Unlimited-OCR [4]. This paper contributes no new model. The contribution is a *measured decision procedure* for combining existing layers over a real, messy, eighteen-year document store, with gates that keep repairs from regressing and keep the evidence quotes honest about their limits.

Limitations worth stating plainly: all fidelity and relevance judgments come from a single frontier vision model, without human raters — and because the judge and the pixel-layer's reader are both vision-language models, correlated failure modes could flatter the pixel-layer numbers specifically; the bake-off and retrieval probes are small stratified samples (15 decks, 37 pages, 12 queries) chosen to stress known failure classes, not to estimate corpus-wide rates; the known-item queries (§2) are author-written — paraphrased from source decks rather than indexed notes, which limits but does not eliminate authorship bias; and for OCR-sourced notes the evidence quotes validate against OCR output rather than source truth (§3.2). These are engineering measurements that guided the build, not benchmark results.

The osQF archive was the testbed, but the pattern is what transfers: layered retrieval with per-document escalation, judged on your own documents rather than on leaderboards, on hardware you control. A side effect we are fond of: the conference's collective memory is now searchable by everyone in the room.

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